The Ephemeris

October 2013
Volume 24 Number 10 - The Official Publication of the San Jose Astronomical Association.

Houge Park October Events

10/6
Solar observing. 2 - 4 p.m.
Fix-It Day 2 - 4 p.m.

10/11
Star party. 7:30 - 10:30 p.m.

10/19
6 - 7:30 p.m. Board of Directors Meeting.
7:30 - 8 p.m. Social time.
General Meeting
8 - 10 p.m. Show and Tell

10/25
Beginner Astronomy Class 7 p.m.
Star party. 7:15 - 10:15 p.m.

10/31
Spooky Star Party
Set your telescope up for kids
Halloween treat!

SJAA Contacts

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Vice President:	Lee Hoglan
Treasurer:	Michael Packer
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Director:	Rich Neuschaefer
Director:	Greg Claytor
Director:	Dave Ittner
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From the Editor - Mina Wagner

Dear readers,

I am sad to report that this will be my last issue as editor of the Ephemeris. I have accepted a new job that, not only will it be full time but it will be very demanding and will need to move away from the little paradise where I live. But this is a very good opportunity for me and am happy and excited to start a new challenging chapter in my life.

I hope to see all of you under the stars now that my 20" mirror has been refigured and re-coated. Unfortunately Mark and I will not be able to make it to the meetings or Houge Park very often but do believe me that I will be thinking of you and missing the good times we had.

I wish the best for you and your loved ones and remember to … keep looking up (as Jack Horkheimer used to say).

David Raimondi, spoke to the SJAA on model rocketry in education. What a blast!

In this issue:

SJAA Yosemite Star Party 2013

By Kevin Lahey

Some of the 2013 Stars Over Yosemite crowd: Gary Mitchell, Jim Van Nuland, Phil Liu, Kevin Lahey, Michael Packer, Paul Mancuso, Kevin Hall, Craig Wandke, unknown, Morris Jones, unknown, unknown

Despite a nearby raging forest fire and an early rising 88% illuminated moon, a large contingent of SJAA members showed up for the Yosemite Star Party, sharing views of the night sky with visitors. We had a variety of scopes, from 76mm reflectors to 5-inch refractors to a 14-inch Dob.

Every year, Yosemite National Park invites several Bay Area astronomy clubs to come up for a weekend to show the night sky to visitors as part of the naturalist program. In exchange for doing this outreach, the park gives us free admission and a place to camp. There's a lottery to determine who gets which weekend; last year, SJAA had a great night during the Perseids, with hardly any moon. This year, we got a big old moon rising early.

When we set up in the amphitheater at Glacier Point, looking out over Half Dome and Yosemite Valley, there were probably 15 different scopes. Lots of us adjusted our finder alignment using the lip of Half Dome just to look through the telescopes at the climbers.

Dome, and quite a few visitors seemed excited midnight; folks seemed to sleep well right up 'til the 6am recycling truck arrived. Earplugs were definitely a win for camp!

As we lounged around on a lazy Saturday afternoon, Michael Packer was kind enough to share his dual-scope solar viewing setup. He's got a hydrogen-alpha for prominences and a Hershel wedge for sunspots. That's definitely a great way to do it, and the contrasting views were fascinating.

Saturday evening was very similar, with the moon arriving slightly later, and a bigger crowd. While most of us left before midnight, Craig Wandke stayed out for several more hours, sketching the moon through his 5-inch Takahashi refractor.

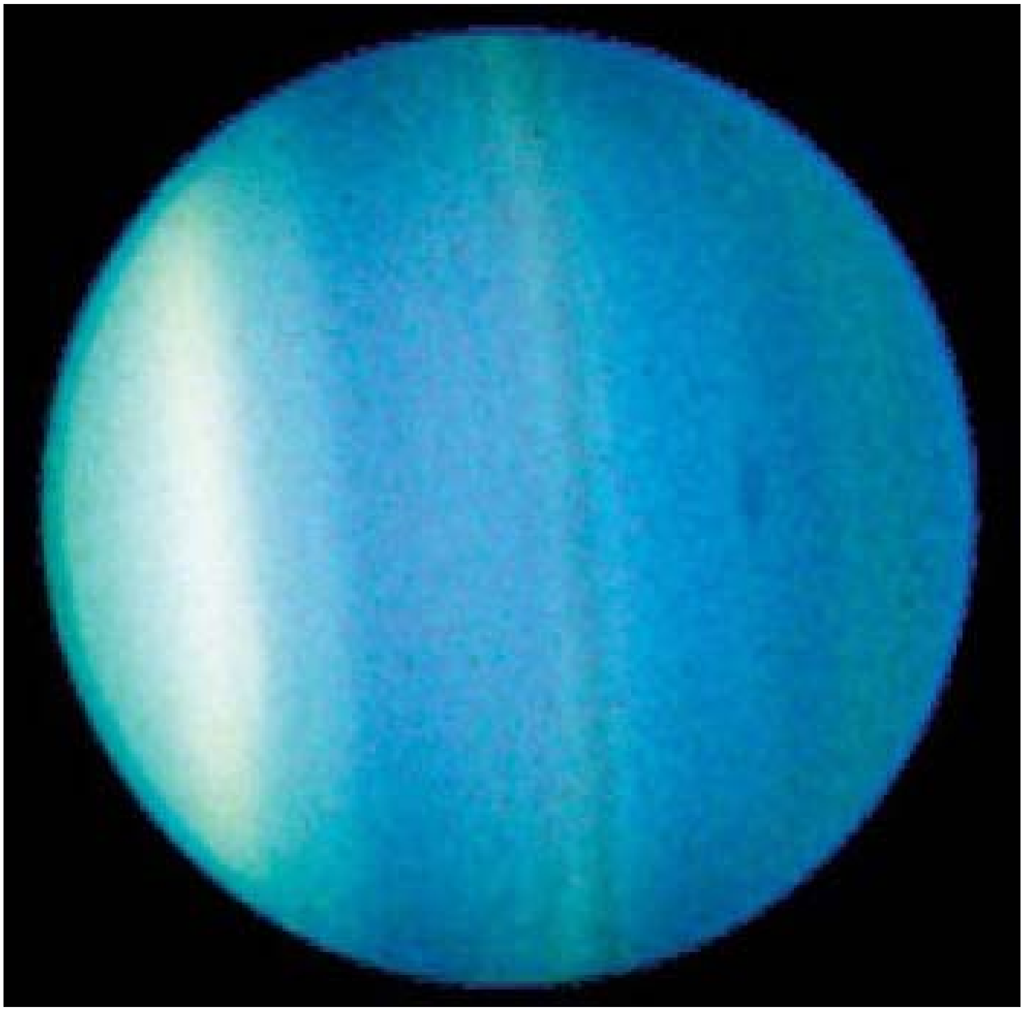
We saw almost no smoke at all from the Rim Fire, which eventually grew to be the third largest in California history (it's burned over 400 square miles so far). Highway 120 and the entrance to the park on 120 were closed, and we could see pyrocumulus clouds to the north, but nothing ever touched us. The campground and the park in general seemed to be pretty empty, though, so the fire must have scared many visitors away. As it was, we were very lucky -- the next weekend, friends reported that ash rained down as far south as Wawona, and smoke was everywhere.

Jim Van Nuland organizes SJAA's Annual Yosemite Star Party

All in all, it was a great trip -- big thanks to Jim Van Nuland, who coordinated everything. If you'd like to participate next year, look for Jim's announcement in the Ephemeris.

Can Amateurs Image Any Detail Of The Cloud Bands On Uranus?

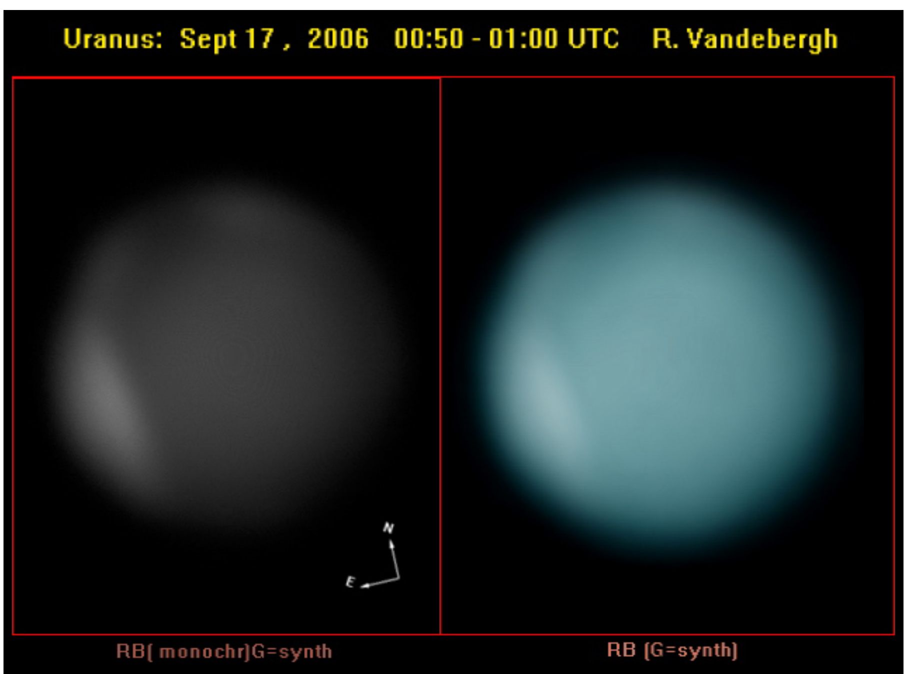
Michael Packer



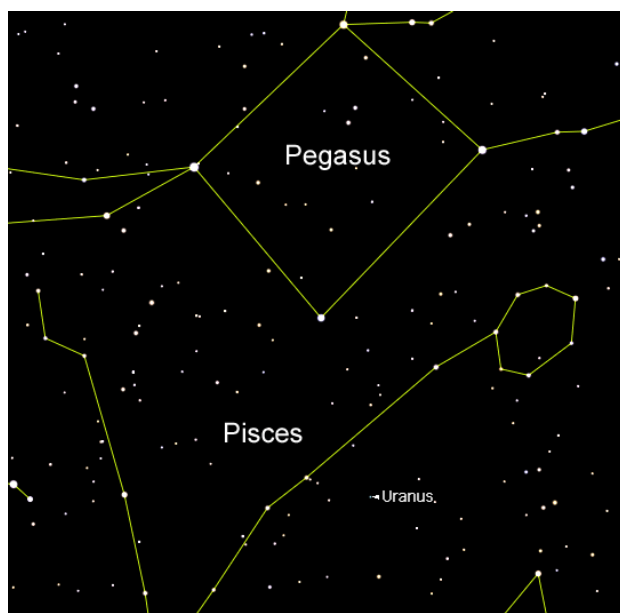
Hubble image of Uranus showing cloud band detail.

Opposition of Uranus is this October 3rd. At 4 Earth diameters Uranus has the third largest planetary diameter in the Solar System. Yet the planet extends on average only 3.5 arc seconds across the sky because of its distance from Sun – 4 times further than Jupiter. Opposition this October will make that 3.7". But 3.7" is still 4 times smaller than the apparent size of Mars (at its next opposition April 2014) or more than 10 times smaller than the size of Jupiter at its January 2014 opposition. In other words the apparent size of Uranus could easily fit inside the Jupiter's Great Red Spot (GRS). For an imager, this means trying to glean detail from an object about 70% the size of the GRS. But amateurs did just that. Ralf Vandebergh (Maastricht Netherlands) took the below image in 2006 clearly showing the NE cloud band and Ed Grafton took some of the earliest successful images in 2004 showing a smack of band detail and along with 5 surrounding moons!

Not too bad! By the Way, I got some of my best views of Mars last opposition when the planet was only 13.89" across. In 2014, we will be a bit closer to the planet and it will subtend 15.16" on April 8th. May 22nd 2016 the elliptical nature of orbits will bring it to us at 18.60" before recedes again. So some great opportunities to see Mar's polar ice caps, strato-cumulus clouds, dust storms, Olympus Mons and more...



Amateur Image of Uranus using a 250mm Newtonian and ATK-1HS Cam-



Uranus is visible all night while at Opposition, and easily visible in any amateur telescope. It is easy to find using the Great Square of Pegasus as a landmark.

Solar Observing at Houge Park

Michael Packer

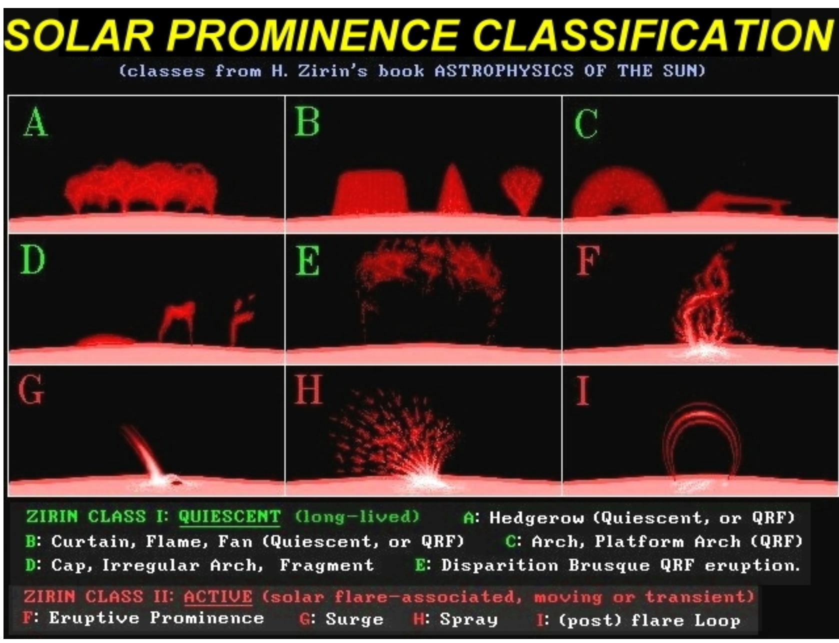
Four Scope Setups were rewarded with a very active view of the Sun. Thanks to Gary Chock for photo.

I expected a less than average turnout this Labor Day weekend. Yet we had 4 Solar Scope Setups and a nice crowd of folks (thanks to all for coming out) that kept the event going past 4:00 o'clock.

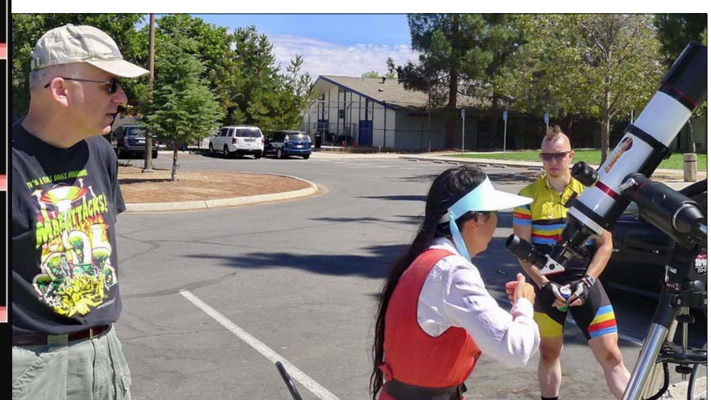
And speaking of that O'clock we noted superb H-Alpha flares (quiescent prominences) at 7, 9 and 12, refractor view, and one "Bic lighter" H-Alpha active flare (surge prominence) at 8 o'clock on the solar dial. The quiescent flares extended some 40,000 miles from the Sun and extended along the Sun 70-80 thousand miles. The narrow surge prominence extended some 80,0000 miles form the surface before we saw it's upper region dissipate.

At it's peak the solar dynamo is sizzling and dazzling. In Kevin Lahey's 10 inch scope we counted 28 sunspots in 4 groups giving a sunspot count of 68. Actually I incorrectly counted four groups instead of three. Had I not done that my estimate would have agreed with the ISN, International Sunspot Number of 58. Below is an image of the Sun for September 1st.

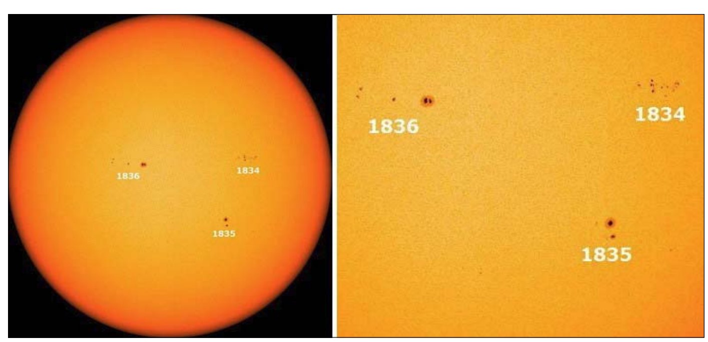
Look forward to seeing everyone next Solar Meetup at Houge October 6th!



Prominence classes – the word flare is a nebulous term. Some prominences are flares in heliophysics sense and generally they all are. Educating kids, I use the word H-Alpha flare instead of the abstract term prominence. This helps young viewers understand what they are looking for while identifying it in the H-Alpha or Chromosphere layer of Sun.



Jenny at 100mm H-Alpha scope checking out the "Bic lighter" prominence.



3 Sunspot Groups and a total Number of 28 spots gives Sunspot Number of (SN = 10G+N) of 58.

Solar Observing at Houge Park

Here is your opportunity to view the Sun safely, in great detail, with knowledgeable and friendly members of the SJAA. Appropriately scheduled the Sunday every month from 2 to 4 p.m. at Houge Park. If you have solar gear and wish to support this program, contact Michael Packer - see the contacts page at sjaa.net

Dark spots, bright spots

Shallow Sky by Akkana Peck

Jupiter rises a little before midnight in October. And on the night of October 11, if you catch it right after it rises at about 10 minutes before midnight, you can catch the end of a nice multiple shadow transit.

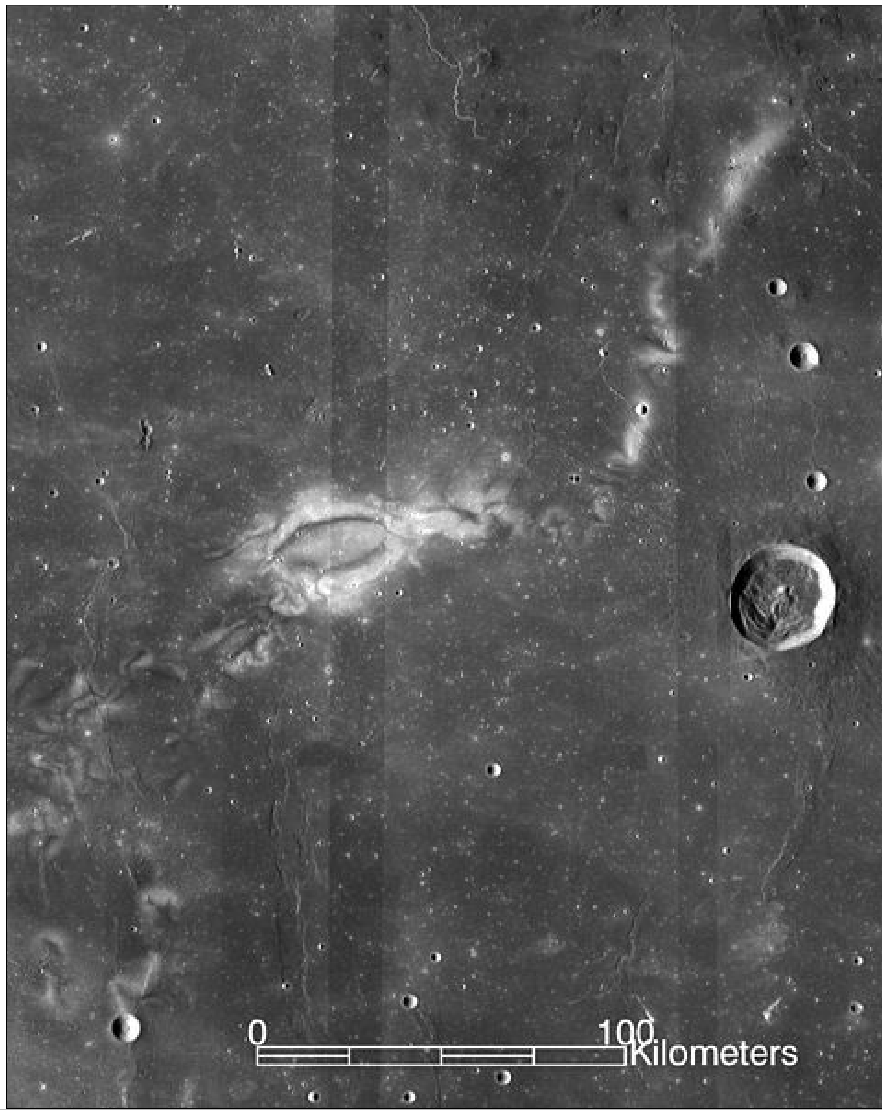
The transit started much earlier, around 7:30 pm, when Callisto's shadow entered the disk -- but that's long before Jupiter rises here in San Jose. Callisto's shadow was joined by Europa's shadow, then Io's, then Io and Europa themselves. By the time Jupiter rises for us, Io and Europa are still in transit across Jupiter's disk, while Io's and Callisto's shadow will be just egressing. Catch it immediately after it rises if you don't want to miss the shadows. The two moons will continue to transit Jupiter until for a bit over an hour, Io leaving the disk before Europa.

The next day, October 12, is this year's fall Astronomy Day. Astronomy Days happen once a season, so four times a year. Which is nice since the sky is so different in different seasons, and you can show off new objects that people haven't seen before.

Uranus is at opposition this month, on October 3. It's in a confusing, star-poor region on the border between Pisces and Cetus. Neptune, too, is well placed, transiting around 10 pm. It's in Aquarius, off the left horn of Capricornus.

You can catch Venus in the evening after sunset. It stays gibbous, right around half phase, all month. Mars is still a morning object. Pluto is still reachable in the early evening, but it's pretty low, well past its prime. You'd have to be pretty dedicated to go after it now Mercury hugs the sun all month.

And then there's the comet. In late October, Comet C/2012 S1 ISON is expected to brightens to magnitude 7 -- bright enough to be visible in binoculars. Unfortunately it's still a morning object, not rising until 3 am, so it's not yet easy for the casual observer. But it's getting easier. October days should also be a good time to look for sunspots. Sunspot Cycle 24 is expected to peak this fall ... though so far this has been the smallest sunspot cycle since February 1906.



Reiner Gamma, an odd and unexplained white pattern near the crater Galileo.

October 18 brings a penumbral lunar eclipse. We'll only see the end of the . eclipse, at moonrise, 6:20 pm, and the eclipse ends about half an hour later. It's not much to get excited about: the effects of a penumbral eclipse, where the moon just barely grazes the brightest part of the Earth's shadow, are so subtle they're almost impossible to see.

But don't take my word for it -- go out and try anyway. It's a good excuse to look at the full moon -- yes, there *are* things to see on the full moon. Full moon is the best time to see albedo features, areas that are only visible because they're brighter or darker than what's around them, like the spectacular rays radiating from Tycho and Copernicus.

Or look for more subtle albedo features like the "fish": Reiner Gamma, an odd and unexplained white pattern near the crater Galileo. I can't decide if it looks more like a fish or a trilobite. Nobody's sure what causes these bright patterns. Reiner Gamma is flat -- it's not a crater or a mountain, and we're not seeing shadows. So what is it?

Well, one clue is that most of the known features of this sort -- there are a handful of others, though most of them are on the moon's far side where we can't see them -- are located 180 degrees away from a major impact basin. For instance, on the far side of the moon opposite Mare Imbrium is a bright spot in Mare Ingenii, and there's another far side bright spot in Mare Marginis, just opposite Mare Orientale. Crisium and Orientale represent two fairly recent, huge wallops. So it seemed that bright and magnetic albedo features might represent something that happened opposite a major impact. However, there's no particularly impressive impact basin opposite Reiner Gamma.

Some have speculated that the odd pattern of Reiner Gamma is due to magnetism. There's a substantial magnetic anomaly in the area, so maybe the pattern is like what you see when you scatter iron filings around a bar magnet. At least if you use a lumpy and somewhat asymmetrical bar magnet.

But for now, the fish -- or trilobite -- remains a mystery. Which makes it all the more interesting to observe.

Quick STARt Program Experience

On September 6th I attended a 3 hour Quick STARt class taught by Dave Ittner at the Houge Park clubhouse in San Jose. He covered many fundamental aspects of astronomy that were enlightening as well as essential for a beginner in amateur astronomy. Topics covered were the 3 type of telescopes used by amateur astronomers as will as the impact of light gathering, magnification, and lenses design to correct color, and much more. He covered common methods of locating heavenly objects including the use of coordinates and sky hopping using constellations. Dave describe the various classifications of stars, galaxies, planets, comets and other celestial bodies and the relationship of size, distance, and color. He explained what we could expect to see with amateur telescopes in terms of detail and how best to view and search for objects in the sky using different magnification of eyepieces. He also encouraged participants to use their naked eyes and a pair of binoculars for viewing the sky.

We then went outside to view the skies with one of the club's telescopes and saw some stars and galaxies. The nigh sky was somewhat obscured by city lights and haze so viewing was less than optimum but for one who has never seen a galaxy that wasn't a photo it was awe inspiring even if the galaxy was just a small smudge through the eyepiece. I wish I could view Saturn but I understand it become visible in the sky around 4-5am.

At the end of the telescope viewing participants returned to the clubhouse where they could check out telescopes and library books to take home.

This class was very interesting and a must for beginners like me who are interested in armature astronomy but know very little. I highly recommend any beginner enroll in this class. I understand you must be a member of SJAA in order to enroll but it is otherwise free and well worth attending.

Frank Geefay – new member and new to amateur astronomy.

Rancho Canada del Oro 9/8 Report

Dave Ittner

Last night (Sunday 9/8) I opened up Rancho Canada del Oro so folks can do some observing under semi dark moonless skies. We had perfect weather – warm and no wind. The lot filled up by 8:30 as we had 15 scopes set up. There were a couple of families that came out and set up their scopes to show their kids the night sky. Rancho continues to attract beginners as folks find out that all of us are very receptive and helpful. Everyone comments on how nice it is to see the band of the Milky Way.

One of the highlights last night was the huge blue-green fireball that streaked across the sky around 11:30 pm. It looked like it had a NE to SW path that took it just above Capricorn. One person commented that it probably was space debris of some type since it was moving so slow.

A number of folks made comments throughout the night that they were seeing some nice meteor streaks. I saw 4 good ones and a few brief flashes. To my knowledge the next major meteor shower is the Draconids with a peak scheduled to happen October 7th-8th. We finally wrapped up around Midnight. Overall it was a darn good turnout for a Sunday night!

A big thanks to all who showed up – a lot of new friends were made and all had a good time.

Milky Way Images from Rancho Canada del Oro

By Zubair Hussain

Above:: Canon 60Da, lens 10-22mm, ISO 1600, aperture wide open, shutter 30 sec. Left: Camera; Canon 60Da, lens 10-22mm, ISO 1600, aperture wide open, shutter 30 sec.

Expedition – Laguna Mountain

Rob Jaworski

The following is my report about our visit to Laguna Mountain, in California's Hernandez Valley, nestled between the coastal and Diablo mountain ranges. Chris Kelly, my friend Tim Berry, a TAC imager named Enrico, and I comprised the entire group that day and night on 07 September 2013. I knew we would be going to a really remote place. My goal was to experience what it would be like to try it from a one night perspective.

Laguna Mountain, managed by United States' Bureau of Land Management (BLM), is a long way to go, but you really get a sense of the central part of California from the drive down there, traveling through Hollister, passing by the east entrance to Pinnacles National Park on highway 25 . The location is just about exactly one hundred miles from base camp Houge Park in San Jose. It takes about two hours with no stops (as measured driving back to SJ in the very early morning).

Highway 25 is a leisurely drive, but does get a bit twisty and a little slow in certain areas. It's well maintained and can be fast in certain stretches. On your way to Laguna, on Highway 25, well past the Pinnacles, you come down off a hill where the road turns sharply to the right. You see a sign pointing to the left toward Laguna Mountain Access, Coalinga Road. This road is not as well maintained, so the going is slower. The pavement is rougher, the road crosses several single lane cattle crossing grates, and low areas, dips, where bridges should be (but are not).

After about 25 minutes on Coalinga Road, you start to approach BLM land, with the Short Fence Trailhead parking area. A little while later, while noticeably gaining elevation, you pass by the Sweetwater Trailhead and campground. A short while after that, you reach your destination, the Laguna Mountain Campground, so you make a right onto the gravel road. It leads up to the campground and terminates at a small clearing where there is a vault toilet building, an information kiosk, and the gate.

At the camping area of Laguna Mountain, there are five sites, on seemingly fresh crushed gravel; it's a little dusty. There are shade structures, about 10 by 12 feet in dimension, on a concrete pad with a metal picnic table bolted, chained to the slab. Sites are decent sized and space away from each other, providing plenty of privacy. There is no water, so be sure to bring plenty of your own. Laguna campsite There are no trash cans, so you have to pack your trash out. There are fire rings but it's hot and dry out there, I would hesitate having a campfire. Site 5 is the best, at the top of the hill, but it's technically behind the gate. To use it, you would have to park in the clearing, near the gate/kiosk and haul all your stuff up the hill, about 30-50 meters around the gate, then back up again to the site. It has good views and is relatively flat, so you could even set up your gear there and observe right at your camp. UPDATE: I'm told that this site is fully available for the asking; contact the local BLM office (links are above in this post) or the SJAA's Rob Jaworski.

The observing area where we set up that night was up the hill beyond the locked gate. To get there, we had to drive about a quarter mile past the gate, for which Chris had the combination as he was also in possession of a permit. Driving up that dirt road, the biggest issue is one fairly steep section of road, maybe fifty feet in length, we had to climb with our vehicles. The 4WD vehicles had no problem but the one non-4WD sedan in our group did have problems climbing. A second run at the hill with more momentum and using a different track, with less loose dirt and rock, proved successful. If the surroundings are wet and muddy, this may be a significant problem as the grade would be impassable, possibly even to 4×4′s.

At the top of a ridge, we set up right in the middle of the well graded dirt road, at N36 21.678′ W120 49.608′ (link to Google map satellite view). Being in the middle of road, if another vehicle had happened to encounter and want to pass us, it would have been impossible unless we all broke down our equipment to let them through. Luckily and expectedly, there was no one trying to get to nowhere beyond to the mountain (or returning from it). (continued on page 8)

(Expedition Laguna Mountain continued)

South wasn't perfect, horizon was slightly obscured by the mountaintop, but IIRC, all of Scorpius' major stars were visible. To the north was the Hernandez Valley, and you could see the reservoir; the horizon is pretty low, the Big Dipper swung all the way down and you can easily see all the stars in Ursa Major's famous asterism. To the northwest, there is a distinct light dome coming from either Hollister or Soledad. Most of us argued for the former while Chris suggested it was the latter. Later, looking at maps, it was apparent that Soledad was the most likely candidate, being much closer and more to the west. Another, much smaller light dome was visible closer to due north, which matches pretty well to where Hollister would be. The hills to the east had a very faint glow coming from behind them, to which I guessed those were probably all the far off towns in the central valley blending together. To the south, there was nothing. In the valley below us, to the west, there were only two, very faint and very far off lights coming from a ranch house or barn. You have to stand up tall and look over the brush to see them. You won't even be able to point at them with most scopes, especially dobs. Other than that, locally, there was nothing else, nothing but occasional coyotes yapping in the distance.

This is more of a site report than an observing report, but I'll make small mentions of what we did up there in the dark. Chris was all over the place with his C14, as usual. Enrico was imaging and showed us versions of the eagle nebula, which looked best in red, though green looked red, and so did blue. Confused? So was I; I don't image. I did manual searching around through the awesome Milky Way, pretending I was an eighteenth century Herschel or Messier, trying to find random things in my XT8. For some reason, I fixated around Delphinus, and found NCG 6834 pretty easily. A lot of my time was also just taking in the dark sky naked eye, what a fabulous place we live, our galaxy. We weren't disappointed with space pebbles crashing into our perfect-for-us atmosphere, many of which seemed to travel from west to east across the southern sky. It just so happened we all saw one come straight down to the south, fiery green, smoke trail and all. We were all using some flavor of Sky Safari all night, a bunch of silhouettes walking around in the dark with dim, red flat panels in our hands. Tim DeBenedictis does a great job of trying to obsolete paper charts, yes he does.

I had obligations the next day, so my buddy Tim and I packed it up and were on the road back to the bay area a little past midnight. We expected to see lots of wildlife on Coalinga Road, then highway 25, but we were all alone. All we saw, the whole way back, was the fuzzy tail of a scampering mammal, as it was disappearing into the roadside brush. I won't mention the small number of nocturns hanging outside the just-closed social establishments of Tres Pinos; they don't count.

Summary: This location is good for people looking for dark skies, and imagers, who don't mind the drive and would be willing to spend the night, driving back the next day. It's not well suited in its current form for any sort of organized event. The location could handle perhaps a dozen observers, perhaps a bit more, scattered throughout the ridge and the campgrounds. I would certainly list it as a place to go and check out with your gear. The land managers are asking for suggestions for improvements and are trying to draw more visitors to come and use their sites in this area. If you like remote, this is the place.

RASC 2014 Handbook

SJAA will be order approximate 50 new versions of the RASC's 2014 Handbooks.

These will be offered to beginners this year, so if you normally purchase one, be sure to watch for announcements.

General Meetings

From Teruo Utsumi

10/19 "Show & Tell"

11/16 Dr. Junwei Zhao - Sun

Quick STARt Program Report

From Dave Ittner

The sessions are filling up to capacity for the most part.

This last session had 11 folks in it. Mainly because I am opening the class up to those who have their own scope (and thus do not need to borrow a club scope). Another reason was that some of these members had either a son, a wife, or friends accompany them.

Advanced Loaner Telescope Program

From Dave Ittner

Things are a little quiet of late. All of the refractors would have been loaned out if we would have had another mount. The club only has 2 mounts yet 3 refractors and 1 SCT. I plan to add another mount by year end.

Jack Z. will soon be ready to start loaning out his 17" custom dob. It is a fine instrument.

Not sure why but there has been no demand for dobs this past month. Maybe everyone who has gone through Quick Start is now wanting to try out a refractor. I have been asked a few times now if the club has a small table top dob or small refractor so folks can try out as a grab n go scope.

SJAA Library Needs Help

From Dave Ittner

Sorry to say that there has been zero progress made in this program over the last month.

Hopefully I will be able to make some time to work on this. I am actively looking for someone who has a passion about the hobby and books to run this program. This could be a major program to assist all of us armchair astronomers enjoy and grow in the

Board Meeting September

The following are selected items from the SJAA's September meeting of the Board of Directors:

Review 2014 Calendar – Dave. Motion was made and all approved to accept the 2014 calendar. Many thanks to JVN for creating it and to Dave for transferring it to the Google Events Calendar.

Recognition Certificates - Greg. Discussion about the need to recognize the members of the club for their service and efforts. Motion was made and all approved that a Recognition/Awards Committee be formed and that Greg be the chairperson.

Finalizing Astronomy Technology Today discount procedure - Ed. Options on how to get the discount code into the hands of members was discussed. It was finally settled on and agreed that the Welcome email would instruct the new/renewing member to email Ed to request the code. Action Item - Rob to modify these emails.

Zeider's 17" Dob – T&Cs review – Dave. Jack Zeider wanted the board to review and approve the Terms and Conditions that he will use. Greg suggested that Jack put in the T&C the repair/replace cost for his scope so that it is fully disclosed up front. SJAA's role is to prequalify folks (1. They have to be members in good standing and 2. They have to be able to star hop and have reasonable experience with Dobsonian telescopes.). Motion made and all approved that the T&Cs look fine.

School Star Party Program

From Jim Van Nuland

Completed events as of Sep. 17, 2013

	Total sched	Good sky	Part. succ.	Cloudy fail	Cancel at noon
Jul	1	-	1	-	-
Start 2013-14 school year					
Aug	2	2	-	-	-
Yosemite NP					
Sep	1	1	-	-	-
Tot	4	3	1	0	0

Scheduled events as of Sep. 17, 2013

	Total	Firm	Working
Sep	1	1	0
Oct	9	7	2
Nov	8	7	1
Dec	4	1	3
Jan	4	2	2
Feb	3	3	0
Mar	1	1	0
Apr	-	-	-
May	1	0	1
Tot	31	22	9

"Working" means that a date has been chosen, but not all approvals have been obtained. With a firm date, some details may remain to be worked out.

A very few events are held at a conference center, county park, etc. Events that piggyback on a regular Houge Park are included if a school, scout, or home school group has requested a star party, and it fits within the capacity of Houge Park.

Fix-It Program Report

From Ed Wong

This past Fixit session took place over the Labor Day weekend so things were kind of slow. We did have a few people stop in with basic questions and we did some basic work on the club's QuickSTARt scopes that were returned

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

San Jose Astronomical Association Membership Form

P.O.Box 28243 San Jose, CA 95159-8243

☐ **New** ☐ **Renewal** (Name only if no corrections)

Membership Type:

☐ Regular — $20
☐ Regular with Sky & Telescope — $53
☐ Junior (under 18) — $10
☐ Junior with Sky & Telescope — $43

Subscribing to Sky & Telescope magazine through the SJAA saves you $5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

☐ **I prefer to get the Ephemeris newsletter in print form (Add $10 to the dues listed on the left). The newsletter is always available online at** http://ephemeris.sjaa.net.

Questions? Send e-mail
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Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA",
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